

**REAL TIME SENSOR BASED MONITORING SYSTEM FOR EARLY DETECTION OF CROP
HEALTH ANOMALIES: A FOCUS ON TARGETED PEST AND DISEASE SURVEILLANCE**

BY

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ENGINEERING IN ELECTRICAL AND ELECTRONICS ENGINEERING**

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OCTOBER, 2025.

DECLARATION

I hereby Declare that this work “**REAL TIME SENSOR BASED MONITORING SYSTEM FOR EARLY DETECTION OF CROP HEALTH ANOMALIES: A FOCUS ON TARGTED PEST AND DISEASE SURVELLIANCE**” is the product of my own research efforts, undertaken under the supervision of **ENGR. DR. UMA UMA** and has not been presented elsewhere for the Award of a degree. All sources have been duly distinguished and appropriately acknowledged.

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CERTIFICATION

This is to certify that the research work for this project titled “**REAL TIME SENSOR BASED MONITORING SYSTEM FOR EARLY DETECTION OF CROP HEALTH ANOMALIES: A FOCUS ON TARGTED PEST AND DISEASE SURVELLIANCE**” and the subsequent preparation of the project by YUSUF BAYO with matriculation number 2020/EN/12734, in the Department of Electrical and Electronic Engineering, were carried out under the supervision of ENGR. DR. UMA UMA, in the Department of Electrical and Electronic Engineering, Alex Ekwueme Federal University, Ndufu-Alike, Ebonyi State.

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DEDICATION

This research work is dedicated to the Almighty God whose mercy and grace led me through
out this journey.

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ABSTRACT

Agriculture is vital for food security and economic stability in developing nations like Nigeria, but it faces significant challenges from pests and diseases that cause 20-40% of global crop losses. Traditional crop monitoring methods are reactive, labour-intensive, and often lead to late detection and excessive use of pesticides, which increases costs and poses environmental and health risks. This study addresses the need for an integrated, affordable, and real-time monitoring system to improve crop protection for smallholder farmers. The research designs a low-cost, sensor-based system that combines environmental data from sensors (temperature, humidity, soil moisture) with visual data from a Raspberry Pi camera. The system uses a four-layered architecture: a sensing layer for data collection, an edge processing layer for on-device analysis, a communication layer for data transmission via GSM, and a user interface layer for alerts and visualization. It employs intelligent algorithms that fuse sensor thresholding with image-based classification using a lightweight Convolutional Neural Network (CNN) to detect anomalies. The system's performance is evaluated based on its accuracy, response time, and reliability in a controlled field environment. This research provides a scalable model for smart farming adoption in Nigeria, promoting resource efficiency and environmental sustainability by enabling targeted interventions. While the study is limited by dataset diversity, hardware computational capacity, and inconsistent power supply, it fills a critical gap in the literature by offering a functional, integrated solution for real-time crop health surveillance in smallholder agricultural contexts.

Keywords: Precision Agriculture, Internet of Things (IoT), Crop Monitoring, Pest and Disease Detection, Machine Learning, Sensor Fusion, Nigeria.

CHAPTER ONE

INTRODUCTION

1.1 Background of the Study

Agriculture is the backbone of food security, economic stability, and social development, particularly in developing nations such as Nigeria. It provides employment for millions, contributes significantly to GDP, and sustains rural communities. Despite its importance, crop production faces persistent challenges, with pests, diseases, and environmental stressors accounting for 20–40% of annual yield losses globally. In Nigeria, outbreaks of pests such as armyworms, stem borers, and locusts, alongside fungal and bacterial diseases, regularly devastate staple crops like maize, rice, cassava, and vegetables.

Traditional monitoring and control strategies rely heavily on manual scouting, farmer experience, or sporadic extension services. These methods are labour-intensive, time-consuming, subjective, and reactive. Farmers often detect outbreaks only when symptoms are visible and damage is already significant, making timely interventions difficult. This delayed response often leads to indiscriminate pesticide application, increasing production costs, causing environmental pollution, and exposing both farmers and consumers to health risks.

The emergence of smart farming technologies offers new opportunities to address these limitations. Internet of Things (IoT) platforms, wireless sensor networks (WSNs), and artificial intelligence (AI) have introduced real-time, data-driven solutions to crop monitoring. Through distributed sensors, farmers can continuously track soil moisture, temperature, humidity, and crop imagery, allowing for early anomaly detection. When combined with machine learning, these systems can predict disease outbreaks, identify pest infestations, and recommend timely interventions.

Recent studies demonstrate that IoT-enabled monitoring improves decision-making, reduces input waste, and enhances resilience against environmental stressors. However, adoption in Nigeria and other developing countries remains limited due to cost, infrastructure gaps, and lack of integrated solutions. This research therefore seeks to design a real-time sensor-based monitoring system for early detection of crop health anomalies, with a particular focus on targeted pest and disease surveillance.

Table 1: Major Crop Pests and Diseases in Nigeria and Their Impact on Yield

Crop	Common Pests/Diseases	Impact on Yield	Reference
Maize	Fall Armyworm (Spodoptera frugiperda), Stem borers, Maize streak virus	20–40% loss if uncontrolled	
Rice	Rice blast (Magnaporthe oryzae), Stem borers, Brown planthopper	Up to 30% loss annually	
Cassava	Cassava mosaic virus, Cassava brown streak disease, Mealybug	25–50% reduction in tuber yield	
Vegetables	Aphids, Whiteflies, Fungal leaf spots, Bacterial blight	Severe quality reduction, up to 35% yield loss	

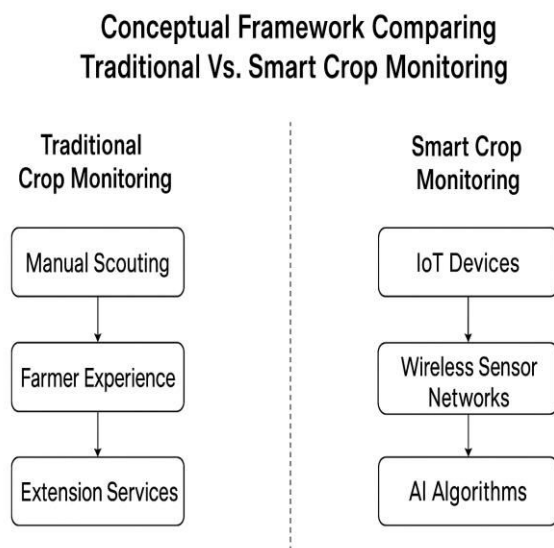


Figure 1: Conceptual Framework Comparing Traditional Vs. Smart Crop Monitoring

1.2 Problem Statement

Despite remarkable progress in agricultural technology, crop health monitoring remains fragmented. Existing systems generally fall into one of two categories:

1. Environmental monitoring systems: These measure soil moisture, humidity, and temperature but fail to provide actionable insights on pest and disease dynamics.
2. Computer vision based systems: These rely solely on leaf imagery for disease classification. While powerful in research settings, they often underperform in real farm environments due to inconsistent lighting, occlusion, and limited datasets.

Because these systems operate in isolation, farmers lack an integrated platform capable of combining environmental data with pest and disease surveillance in real time. This limitation results in delayed detection of crop anomalies, excessive pesticide use, reduced yields, and environmental degradation.

In Nigeria, the absence of affordable, real-time, sensor-based systems has left farmers highly vulnerable to sudden pest outbreaks such as the Fall Armyworm invasion of maize fields in 2016–2017 and recurring plant diseases such as cassava mosaic virus and rice blast. These challenges undermine the agricultural sector's ability to meet growing food demands.

There is therefore a pressing need for an integrated, low-cost, sensor-based monitoring system that can simultaneously track environmental parameters, detect pest/disease anomalies, and issue real-time alerts.

1.3 Aim and Objectives of the Study

Aim:

The aim of this research is to design and implement a real-time sensor-based monitoring system for early detection of crop health anomalies, with emphasis on targeted pest and disease surveillance.

Specific Objectives:

To achieve the aim, the study will:

1. Develop a sensor-based system that integrates environmental monitoring (temperature, humidity, soil moisture) with visual data collection for crop health assessment.
2. Design and implement intelligent algorithms capable of detecting and classifying pest and disease anomalies in real time.
3. Evaluate the system's performance in terms of detection accuracy, response time, and reliability under field conditions.

1.4 Significance of the Study

This study is significant in several ways. First, it provides farmers with a real-time decision-support tool that enables early detection of crop health anomalies, thereby reducing losses caused by pests and diseases.

Second, the system promotes resource efficiency by encouraging targeted pesticide application. This reduces production costs for farmers, minimizes the environmental hazards associated with excessive chemical use, and aligns with global efforts toward sustainable agricultural practices.

Third, the study contributes to the advancement of smart farming technologies in Nigeria and other developing regions. It demonstrates how low-cost sensors, artificial intelligence, and wireless communication can be integrated into a functional prototype, offering a scalable model for broader agricultural adoption.

Finally, the research provides an academic contribution by addressing a critical gap in literature: the lack of integrated systems that combine environmental monitoring with real-time pest and disease surveillance.

1.5 Scope of the Study

This study is restricted to the design and prototype development of a real-time sensor-based monitoring system for crop health surveillance. The system focuses on integrating environmental sensing (temperature, humidity, soil moisture) with image-based detection of pest and disease symptoms.

The research is limited to selected crops grown under controlled or small-scale field conditions, where the prototype can be tested and validated. The hardware implementation makes use of low-cost sensors, microcontrollers, and imaging modules, while the software

components utilize lightweight machine learning algorithms for anomaly detection and alert generation.

The scope does not extend to large-scale deployment across multiple agro-ecological zones, commercial farm integration, or full-scale economic evaluation. Instead, the emphasis is on demonstrating the technical feasibility of the system and assessing its performance metrics namely, detection accuracy, responsiveness, and reliability.

1.6 Limitations of the Study

Like any research, this study is subject to certain limitations that may affect its scope and outcomes. First, the availability of diverse datasets for training the pest and disease detection algorithms is limited. Many existing datasets, such as PlantVillage, are collected under controlled conditions and may not fully reflect real-world variability in lighting, background, and crop diversity.

Second, the hardware resources used in the prototype, such as low-cost sensors and microcontrollers, have limited precision and computational capacity. This may restrict the complexity of algorithms that can be deployed on the edge device and influence the overall accuracy of detection.

Third, power supply and network reliability pose practical challenges, especially in rural areas of Nigeria where electricity and internet access are inconsistent.

Finally, financial and time constraints limit the extent of field trials to a small-scale prototype rather than large-scale implementation. Despite these challenges, the study provides valuable insights and establishes a foundation for future improvements and broader deployments.

CHAPTER TWO

LITERATURE REVIEW

2.1 Concept of Precision Agriculture

Agriculture has traditionally been practiced in a generalized manner, where inputs such as fertilizer, pesticides, and irrigation water are applied uniformly across an entire farm. While this approach may be simple, it often leads to inefficiencies because not all parts of a field have the same soil fertility, moisture level, pest pressure, or crop health. Precision agriculture (PA) emerged as a response to this problem, providing a framework for site-specific crop management (SSCM) where farming decisions are tailored to the unique conditions of different zones within a farm .

At its core, precision agriculture is built on data acquisition, analysis, and decision-making. Data is collected through tools such as geographic information systems (GIS), global positioning systems (GPS), drones, satellites, unmanned aerial vehicles (UAVs), soil and environmental sensors, and smart farm machinery. Once collected, the data is analyzed often with the help of artificial intelligence (AI) and machine learning (ML) algorithms to generate actionable insights for farmers. These insights guide decisions such as when and where to irrigate, how much fertilizer to apply, or whether pest control is necessary in a specific zone of the farm.

2.1.1 Precision Agriculture

Precision Agriculture (PA), also known as site-specific crop management (SSCM), is a farming approach that leverages information technology to optimize the management of inputs such as water, fertilizer, pesticides, and labor. Unlike traditional farming that applies uniform treatments across entire fields, PA recognizes the variability in soil fertility, crop

health, and pest pressure. By adopting data-driven methods, PA enables farmers to treat each plot or even individual plants based on their specific needs.

2.1.2 Core Components of Precision Agriculture

The PA framework is typically built on three pillars:

1. Data Acquisition:

Sensors, drones, satellites, and automated machinery collect large volumes of environmental and biological data. For example, soil sensors provide real-time information about moisture and pH, while drones capture crop canopy imagery.

2. Data Analysis:

Collected data is processed using AI, machine learning, or GIS-based models to detect patterns. For instance, ML algorithms can predict pest outbreaks from environmental data, or identify nutrient deficiencies from crop imagery.

3. Decision-Making and Action:

The insights generated are translated into variable-rate actions. This includes targeted fertilizer application, precision irrigation, and selective pesticide spraying.

2.1.3 Benefits of Precision Agriculture

The benefits of PA are multi-dimensional:

Increased Productivity: Optimized use of resources leads to higher and more stable yields.

Cost Efficiency: Reduces overuse of inputs such as water, fertilizer, and pesticides, lowering production costs.

Environmental Sustainability: Minimizes agrochemical runoff and soil degradation, aligning with sustainable farming practices.

Risk Management: Enables timely interventions against pests and diseases, reducing yield losses.

Data-Driven Planning: Historical records assist in planning crop cycles and predicting future outcomes.

2.1.4 Global Adoption of Precision Agriculture

In developed regions, such as North America and Europe, PA is widely adopted due to access to advanced technologies. Farmers in the United States use GPS-guided tractors, UAV-based scouting, and multispectral sensors to manage large fields. Similarly, the European Union promotes PA through sustainability initiatives, offering subsidies for digital farming tools.

In developing regions, adoption is limited. Smallholder farmers in sub-Saharan Africa face barriers including high technology costs, poor electricity infrastructure, and limited technical expertise . However, innovations in low-cost IoT sensors, mobile phone apps, and solar-powered devices are gradually making PA accessible.

2.1.5 Precision Agriculture in Nigeria

Agriculture accounts for nearly 23–25% of Nigeria’s GDP and employs about 70% of the population. However, the sector suffers from low productivity due to pest outbreaks, poor irrigation practices, and climate variability. Traditional uniform-input farming practices often lead to resource wastage and poor yields.

Recently, Nigerian universities, research centers, and start-ups have initiated PA projects. Examples include IoT-enabled irrigation systems, mobile-based advisory apps, and UAV-based crop monitoring for maize and cassava . Although these solutions remain in pilot stages,

they highlight the potential of PA in Nigeria if cost, scalability, and infrastructure issues are addressed.

2.1.6 Challenges and Future Directions

Despite its benefits, PA adoption faces obstacles such as:

High Initial Cost of advanced sensors and drones.

Technical Complexity requiring farmer training.

Poor Infrastructure (electricity and internet connectivity).

Data Overload, where farmers struggle to interpret complex datasets.

Future trends include affordable open-source platforms, lightweight AI for mobile apps, and solar-powered IoT systems designed for rural farms.

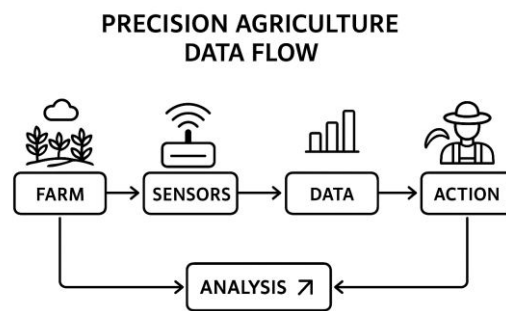


Figure 2. 1 : Precision Agriculture Data Flow

2.2 IoT-Based Agricultural Monitoring Systems

The Internet of Things (IoT) has become a central pillar of modern agricultural monitoring by enabling distributed sensing, automated data collection, and remote actuation for farm management. IoT agricultural systems typically consist of (a) field sensors to measure environmental and crop parameters, (b) short-/long-range communication links (e.g., ZigBee,

LoRa, NB-IoT, Wi-Fi), (c) edge or gateway devices for preprocessing, and (d) cloud or on-premises servers for storage, analytics, and user interfaces. This layered architecture supports continuous, near-real-time visibility into field conditions and enables automated control actions such as irrigation or targeted spraying.

Sensor types used in IoT farming range from basic environmental probes (temperature, relative humidity, soil moisture/temperature, electrical conductivity) to more advanced units (spectral sensors, leaf wetness, gas sensors for volatile organic compounds) and imaging devices (RGB, multispectral, thermal cameras). Wireless Sensor Network (WSN) research has emphasized sensor selection, placement strategies, duty cycling, and energy-aware sampling to balance data fidelity with lifetime and cost constraints issues particularly important for battery- or solar-powered deployments. Communication technologies are chosen according to range, data rate, and power budget. Short-range protocols (ZigBee, BLE) are suitable for dense cluster deployments, while Low Power Wide Area Network (LPWAN) technologies such as LoRa/LoRaWAN and NB-IoT are increasingly preferred for distributed smallholder landscapes because they extend coverage and dramatically reduce per-node energy consumption. Several recent case studies and reviews document LoRaWAN's utility for field monitoring and low-cost automatic irrigation control, noting trade-offs in throughput, latency, and duty-cycle regulatory limits.

Edge computing has emerged as a practical solution to the latency, bandwidth, and privacy problems of cloud-only IoT systems. By performing preprocessing (noise filtering, compression), lightweight analytics, and even on-device inference (pruned CNNs, MobileNet variants) at gateways or SBCs (e.g., Raspberry Pi, Jetson Nano), systems can reduce cloud traffic and provide faster alerts for time-critical events. Hybrid edge-cloud architectures where edge nodes handle real-time detection and the cloud performs heavier model training, aggregation, and dashboards are now a recommended pattern for agricultural IoT applications.

Data management and analytics in IoT agriculture combine time-series sensor processing, image analytics, and often ML models for anomaly detection and prediction. Recent systematic reviews show that combining environmental data with image-based models improves detection robustness compared with single-modality approaches, while also supporting predictive tasks such as irrigation scheduling and disease risk forecasting. However, model generalizability is limited by data heterogeneity (different crops, lighting, camera angles) and the need for labeled, locally representative datasets.

Practical benefits demonstrated in deployments include improved irrigation efficiency, earlier detection of pest/disease outbreaks, and reduced input use. For example, IoT-driven irrigation controllers have delivered measurable water savings and yield stabilization in empirical studies, and cloud-enabled sensor platforms have enabled automated greenhouse control and remote monitoring for smallholder tomato production. Nevertheless, several operational challenges persist power reliability, sensor drift and calibration, network availability in rural areas, upfront cost, interoperability, and data governance/privacy concerns that affect adoption and long-term sustainability.

2.2.1 IoT Architecture in Agriculture

An IoT-based agricultural system typically consists of four layers:

1. Perception Layer: Sensors for soil, crop, and environmental monitoring.
2. Network Layer: Wireless protocols like ZigBee, LoRa, NB-IoT, GSM.
3. Middleware Layer: Platforms for data integration and analytics (cloud or edge computing).
4. Application Layer: Farmer-facing dashboards, SMS alerts, or mobile apps.

2.2.2 Communication Protocols in Agricultural IoT

Different communication protocols have trade-offs:

Bluetooth/Wi-Fi: Short-range, high data rates, high power consumption.

ZigBee: Short-to-medium range, low power, mesh networking.

LoRa: Long range (up to 15 km), low power, ideal for small packets.

NB-IoT: Reliable, long range, requires telecom infrastructure.

GSM/3G/4G: Widely available in Nigeria, but energy-hungry, costly for sensors.

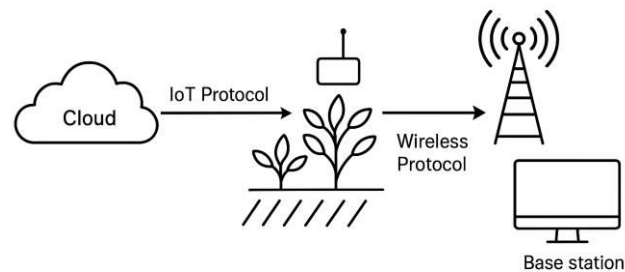


Figure 2. 2 : Precision Agriculture Data Flow

Table 2: Comparative Analysis of IoT Communication Protocols

Protocol	Range	Data Rate	Power Consumption	Suitability for Agriculture
Bluetooth/Wi-Fi	10–100 m	High	High	Short-range, high-data monitoring
ZigBee	10–100 m	Low–Medium	Low	Greenhouses, mesh networking

LoRa	2–15 km	Low	Very Low	Wide farms, rural areas
NB-IoT	1–10 km	Low– Medium	Low	Areas with telecom infrastructure
GSM/3G/4G	1–20 km	Medium– High	High	Widely available, but energy-costly

2.2.3 Advantages and Challenges of IoT in Agriculture

Advantages: Real-time monitoring, resource efficiency, anomaly detection, decision-making.

Challenges: Power supply, internet gaps, cost, and farmer training.

2.3 Sensor Technologies for Crop Health Monitoring

Sensors are the backbone of modern agricultural monitoring systems because they provide the raw data needed for intelligent decision-making. By continuously measuring environmental and crop-related variables, sensors enable farmers to detect subtle changes in field conditions that may signal stress, pest outbreaks, or disease onset. Broadly, sensors used in agriculture fall into three categories: environmental sensors, soil sensors, and imaging sensors.

2.3.1 Environmental Sensors

Used for temperature, humidity, radiation, and leaf wetness important for predicting fungal outbreaks such as rice blast.

2.3.2 Soil Sensors

Measure soil moisture, nutrient availability, and pH. Help optimize irrigation and fertilizer application.

2.3.3 Imaging Sensors

RGB cameras: Cheap, widely used for leaf disease apps.

Multispectral cameras: Used for NDVI vegetation indices.

Hyperspectral cameras: Detect invisible symptoms, but costly.

Thermal cameras: Identify water stress by canopy temperature.

2.3.4 Sensor Fusion

Combining multiple sensors reduces false positives and improves anomaly detection.

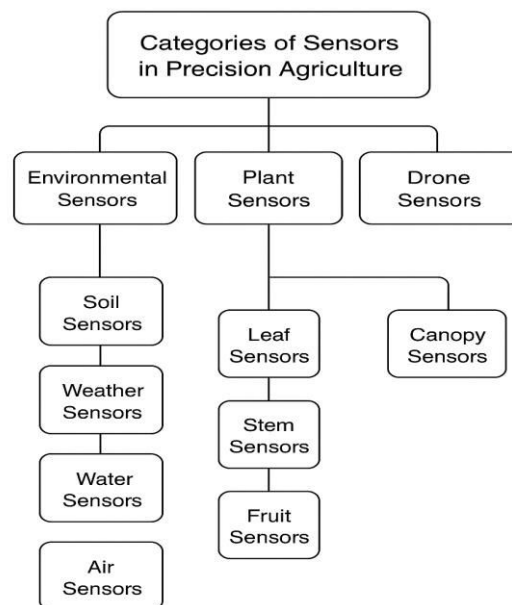


Figure 2. 3: Categories of Sensors in Precision Agriculture

Table 3: Summary of Sensor Types for Crop Monitoring

Category	Examples	Applications / Pests & Diseases	Impact / Advantages	Limitations
Maize	Fall Armyworm (<i>Spodoptera frugiperda</i>), Stem borers, Maize streak virus	Pest & disease infestation	20–40% yield loss if uncontrolled	Needs timely control measures
Rice	Rice blast (<i>Magnaporthe oryzae</i>), Stem borers, Brown planthopper	Pest & disease infestation	Up to 30% annual yield loss	Intensive management required
Cassava	Cassava mosaic virus, Cassava brown streak disease, Mealybug	Pest & disease infestation	25–50% tuber yield reduction	Resistant varieties not always available
Vegetables	Aphids, Whiteflies, Fungal leaf spots, Bacterial blight	Pest & disease infestation	Quality reduction, up to 35% yield loss	Sensitive to environment
Environmental	Temperature,	Microclimate	Low-cost,	Placement

Sensors	humidity, leaf wetness	monitoring & disease prediction	predictive potential	affects accuracy
Soil Sensors	Moisture probes, pH sensors	Irrigation scheduling, soil fertility management	Direct soil insights	Calibration requirements, cost issues
Imaging Sensors	RGB, multispectral, hyperspectral, thermal	Disease & stress detection	Visible & invisible symptom detection	High cost, weather sensitive
Sensor Fusion	Environmental + soil + imaging	Integrated anomaly detection, robust decision support	Higher accuracy, reliable predictions	Complexity & high computational demand

2.4 Machine Learning and AI in Pest and Disease Detection

2.4.1 Deep Learning Approaches

CNNs have shown state-of-the-art accuracy (>95%) in controlled datasets (e.g., PlantVillage) . Advanced models such as YOLO and Faster R-CNN also provide real-time localization of infected leaves.

2.4.2 Hybrid and Multimodal AI

Combining imaging with environmental data improves robustness. For example, Li *et al.* (2020) integrated RGB and humidity data to improve disease detection.

2.4.3 Challenges

AI systems face dataset bias, limited computation on low-cost devices, and lack of explainability, which reduces farmer trust.

AI Workflow for Pest and Disease Detection

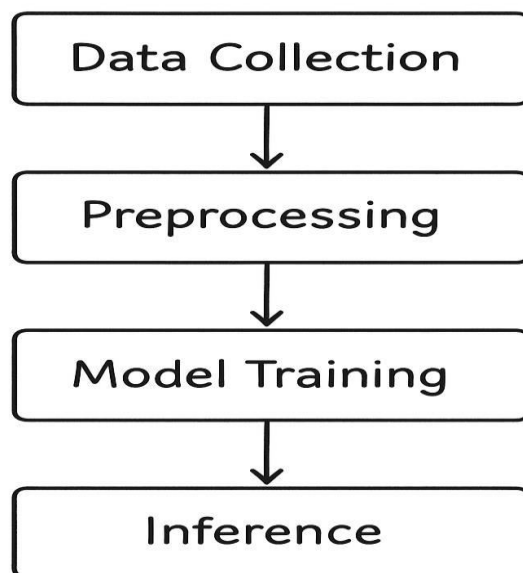


Figure 2. 4: AI Workflow for Pest the and Disease Detection.

2.5 Review of Related Works

2.5.1 IoT and Sensor-Based Studies

Patil & Kale (2016): WSN for soil/environment monitoring, but no pest detection.

Adeyemi *et al.* (2020): Low-cost IoT device for Nigerian farms, but lacked AI integration.

2.5.2 Imaging and Computer Vision Studies

Mohanty *et al.* (2016): CNNs achieved >95% accuracy on PlantVillage.

Ferentinos (2018): CNNs on field data, accuracy dropped to ~85% due to variability.

2.5.3 Hybrid Approaches

Li *et al.* (2020): Multimodal AI outperformed image-only models.

Shamshiri *et al.* (2021): Greenhouse IoT + vision, but limited scalability.

2.6 Summary of Literature Review

This chapter has reviewed precision agriculture, IoT architectures, communication protocols, sensor technologies, and AI-based pest/disease detection. From the literature, it is evident that:

1. IoT and sensor technologies provide real-time monitoring, but most systems are fragmented.
2. AI models achieve high accuracy in controlled datasets, but real-world generalization remains limited.
3. Few systems integrate environmental and imaging data for holistic crop monitoring.
4. Cost and infrastructure constraints hinder adoption in developing regions such as Nigeria.

The identified gap is the lack of affordable, integrated, real-time monitoring systems for pest and disease surveillance in smallholder contexts. This study addresses that gap by proposing a low-cost IoT + AI-based framework for Nigerian agriculture.

CHAPTER THREE

METHODOLOGY

3.1 Hardware Components (Sensors, Micro Controllers)

3.1 Hardware Components

The hardware components form the physical foundation of the monitoring system, supporting data collection, communication, and control. At the center of the setup is the Raspberry Pi 4, which serves as the main controller due to its quad-core processor, in-built Wi-Fi, and compatibility with modern AI libraries that enable efficient sensor data processing and computer vision operations. The Pi Camera Module V3 captures high-resolution images of the tomato crops on the artificial farmland, providing input for detecting discoloration, pest attacks, and disease spots. Its LED and infrared lighting ensure clear image capture even in low-light or nighttime conditions. The system's Wi-Fi connectivity enables real-time data transmission and remote communication with a local server or cloud platform, while a micro SD card serves as the primary storage for the operating system, sensor logs, AI datasets, and trained models. Additional environmental sensors—including temperature, humidity, pest-detection, and crop-specific modules—supply essential background data for identifying anomalies and predicting pest or disease activity. The entire setup is powered by a solar energy unit with a battery backup, ensuring uninterrupted operation during power outages. Together, these components create a reliable, scalable, and sustainable hardware framework for early crop health monitoring. Below are the table show the component and its Justification.

Table 3.1: Hardware Components Used in the Monitoring System

Component	Function	Justification for Use
Raspberry Pi 4	Central processing and control	Supports AI and real-time monitoring
Pi Camera Module V3	Captures crop images for analysis	High resolution, AI-compatible
LED/IR Light	Night-time image support	Enables continuous 24/7 monitoring

Micro SD Card	Storage for OS and AI models	Low cost, portable, and expandable
Sensors (Temp, Humidity, Pest, Imported)	Environmental monitoring	Provide critical data for anomaly forecasting
Drone/Farmland Platform	Aerial or ground-based deployment	Expands field coverage and scalability
Power Source	Solar + Battery backup	Ensures sustainability in off-grid conditions

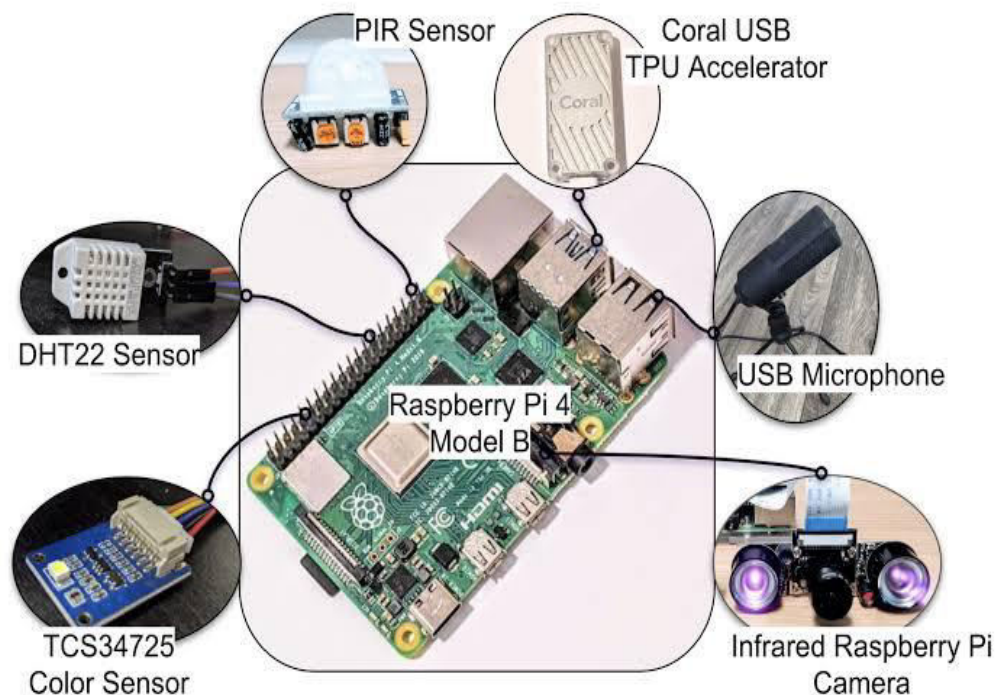


Figure 3. 1: Hardware Connection Setup Showing All Components

3.2 Software Components (Ai Models, Data Processing Tools, Interfaces)

The intelligence layer of the monitoring system is designed around Python-based integration of sensors, camera, and artificial intelligence modules. Python serves as the main programming language due to its broad support for machine learning and hardware control libraries. Using Python, scripts were developed on the Raspberry Pi to collect environmental data from humidity, temperature, and soil moisture sensors. These data streams are processed

in real-time and transmitted to the monitoring interface for visualization and decision-making. In addition to the sensors, a Pi Camera module was installed to continuously capture images of potato leaves. These images undergo preprocessing through the OpenCV library, which handles operations such as resizing, segmentation, and feature enhancement to improve model accuracy. A Convolutional Neural Network (CNN) model, trained on three classes of potato leaves (healthy, early blight, and late blight), was implemented using TensorFlow. The trained model is deployed locally on the Raspberry Pi to enable on-edge anomaly detection without continuous internet dependence. This ensures fast and efficient identification of leaf diseases directly at the farm site. The processed sensor readings and CNN classification results are then transmitted to a user dashboard built on the Blynk platform. Blynk provides a mobile-based interface that allows real-time monitoring of sensor data and displays image classification outputs directly on the screen. The system also leverages HTTP communication for smooth data exchange between the local device and the dashboard. Collectively, these software tools form an intelligent and interactive architecture, allowing farmers to observe environmental parameters, detect plant anomalies instantly, and take corrective actions through a single, connected platform.

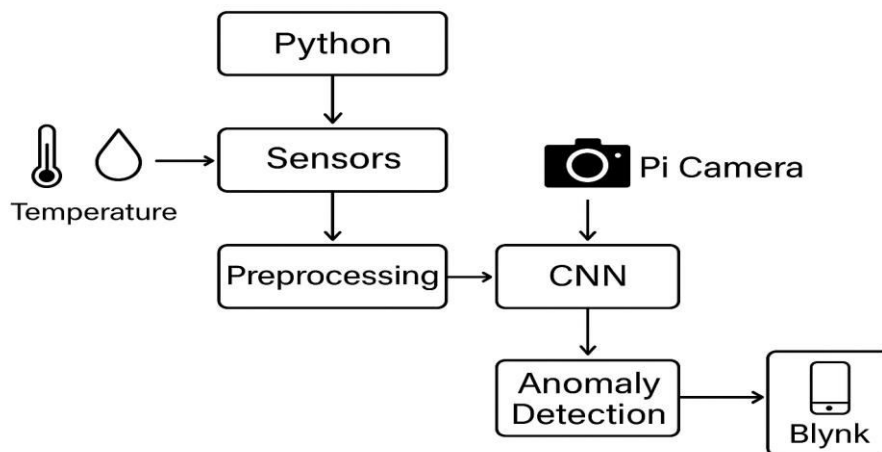


Figure 3. 2: Software architecture for sensor and AI-based detection

3.3 General Methodology Employed

The methodology outlines the approach used in developing the intelligent monitoring system for detecting crop anomalies in real-time. It involves configuring the Raspberry Pi, installing required software libraries, acquiring sensor and image data, training a CNN model, and visualizing outputs through the Blynk dashboard. Each phase contributes to an integrated system capable of reliable and continuous crop health monitoring. The workflow begins with Raspberry Pi setup, sensor and camera installation, library configuration, CNN model deployment, and dashboard integration. The following subsections describe these processes sequentially.

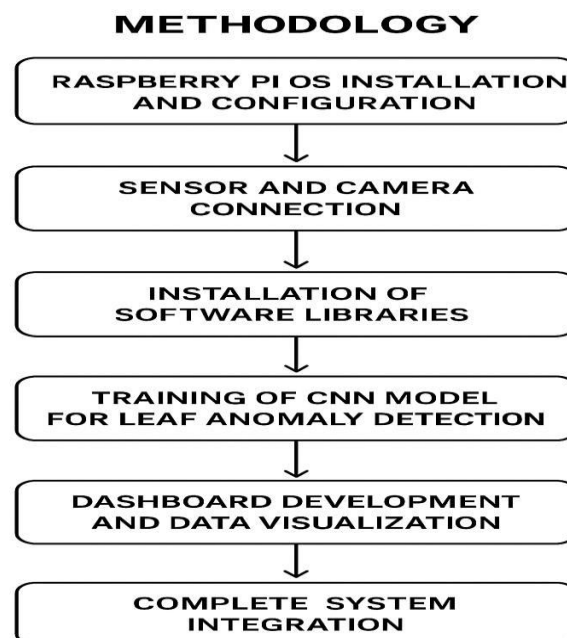


Figure 3. 3: General methodological workflow for real-time crop health anomaly detection.

3.3.1 Raspberry Pi Os Installation And Configuration

The first stage involves setting up the Raspberry Pi 4 as the control unit. Raspberry Pi OS is installed on a microSD card and configured for network access, SSH, and VNC. The Pi is updated and connected to the internet, enabling smooth interaction with peripheral devices and remote monitoring. GPIO pins are then enabled for interfacing with sensors and the Pi

Camera module. This configuration establishes the system base for data collection and image capture. It ensures that the Raspberry Pi can process both sensor readings and camera inputs simultaneously without lag, forming the foundation for subsequent stages.

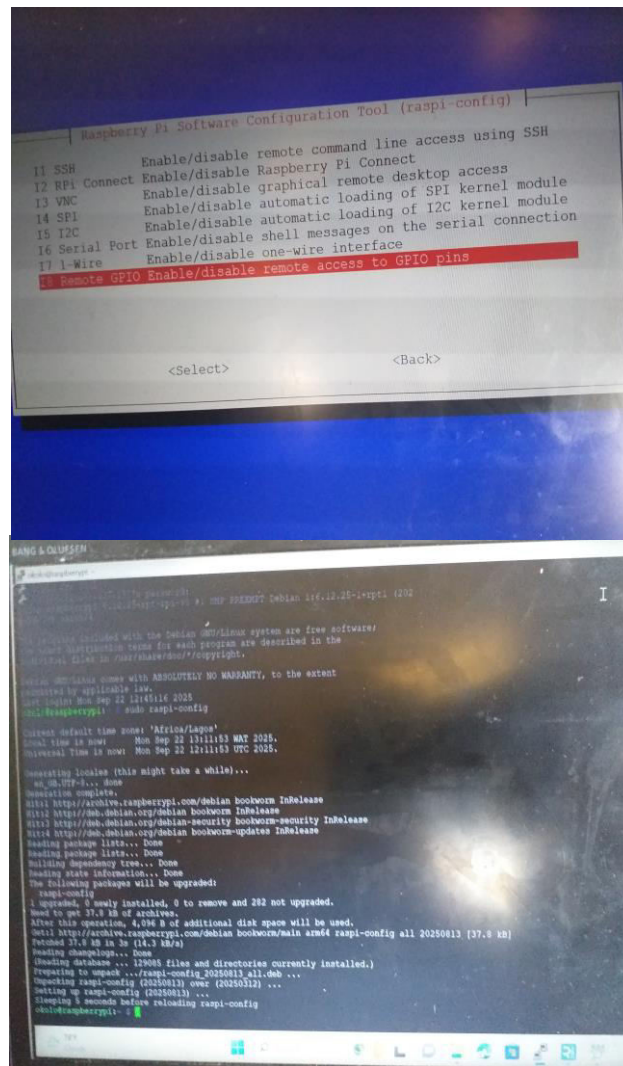


Figure 3. 4: Setup of Raspberry Pi OS and configuration for sensor and camera communication.

3.3.2 Sensors And Camera Connection

Humidity, temperature, and soil moisture sensors are connected to the Raspberry Pi via GPIO pins. Each sensor is calibrated to ensure accurate readings. The Pi Camera Module V3 is mounted above the crops to capture clear leaf images at regular intervals. These devices continuously collect environmental and visual data. The synchronized operation between

sensors and camera enables a consistent dataset used to analyze crop health and detect visible signs of pest or disease anomalies.

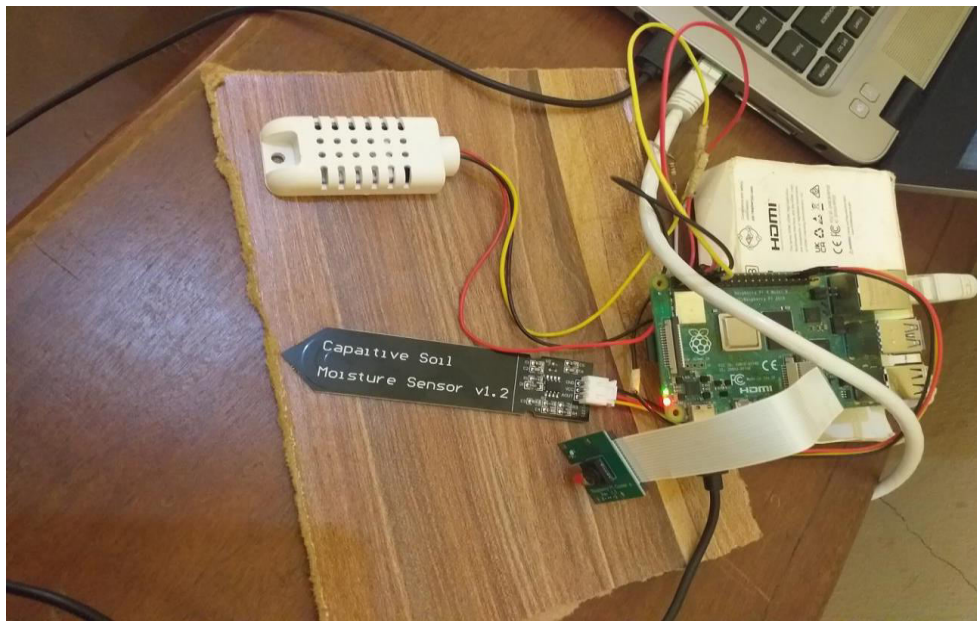


Figure 3. 5: Sensor and camera connection layout for field data acquisition.

3.3.3 Installation Of Software Libraries

Python is used as the main programming language for its flexibility and AI compatibility. Essential libraries such as Adafruit_DHT, OpenCV, TensorFlow, and Pandas are installed to support data acquisition, image preprocessing, model training, and dataset management. OpenCV handles image resizing, segmentation, and enhancement, while TensorFlow is used for CNN training and inference. Pandas organizes both sensor and image data for efficient processing, ensuring seamless hardware–software interaction on the Raspberry Pi.

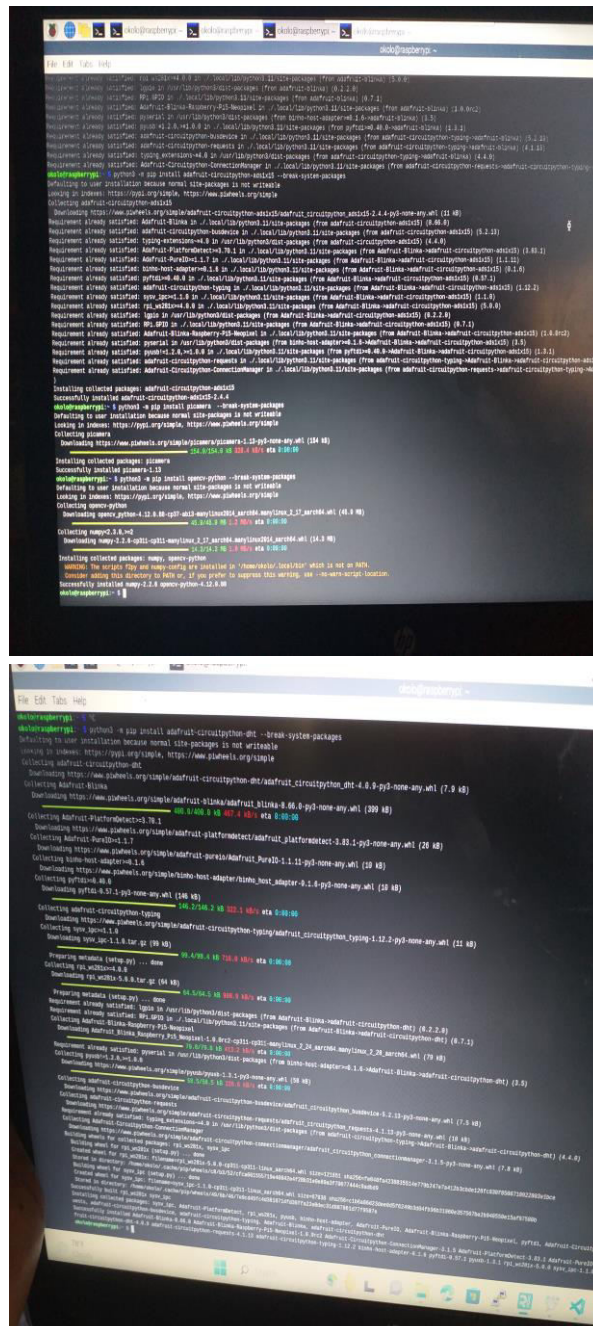


Figure 3. 6: Library and software architecture for system functionality.

3.3.4 Training Of Cnn Model For Leaf Anomaly Detection

A Convolutional Neural Network (CNN) model is developed to classify three leaf categories—healthy, early blight, and late blight. Using TensorFlow, the model learns image features such as color variation, texture, and lesion shapes for accurate classification. After validation, the trained model is converted to TensorFlow Lite format for deployment on the

Raspberry Pi. This allows real-time prediction on captured images without depending on constant internet connectivity, improving speed and efficiency.

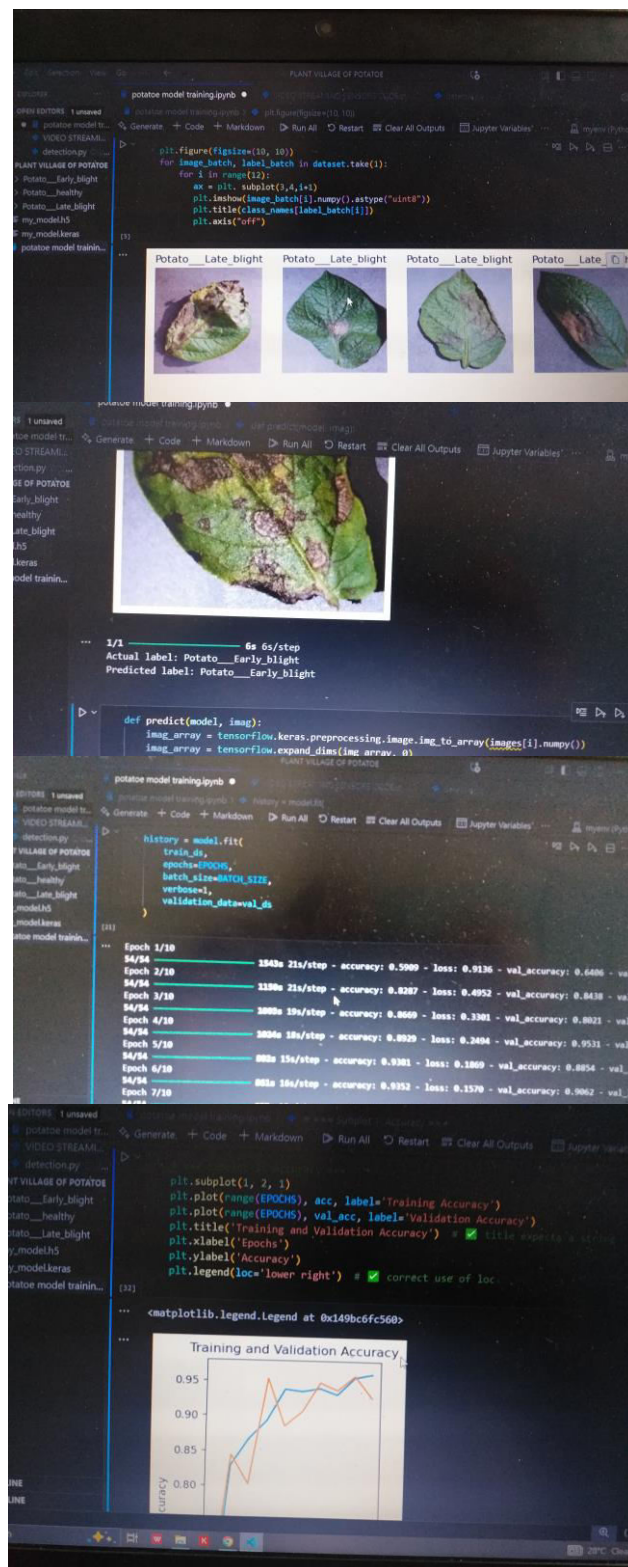


Figure 3. 7: Workflow of CNN model training and conversion to TensorFlow Lite.

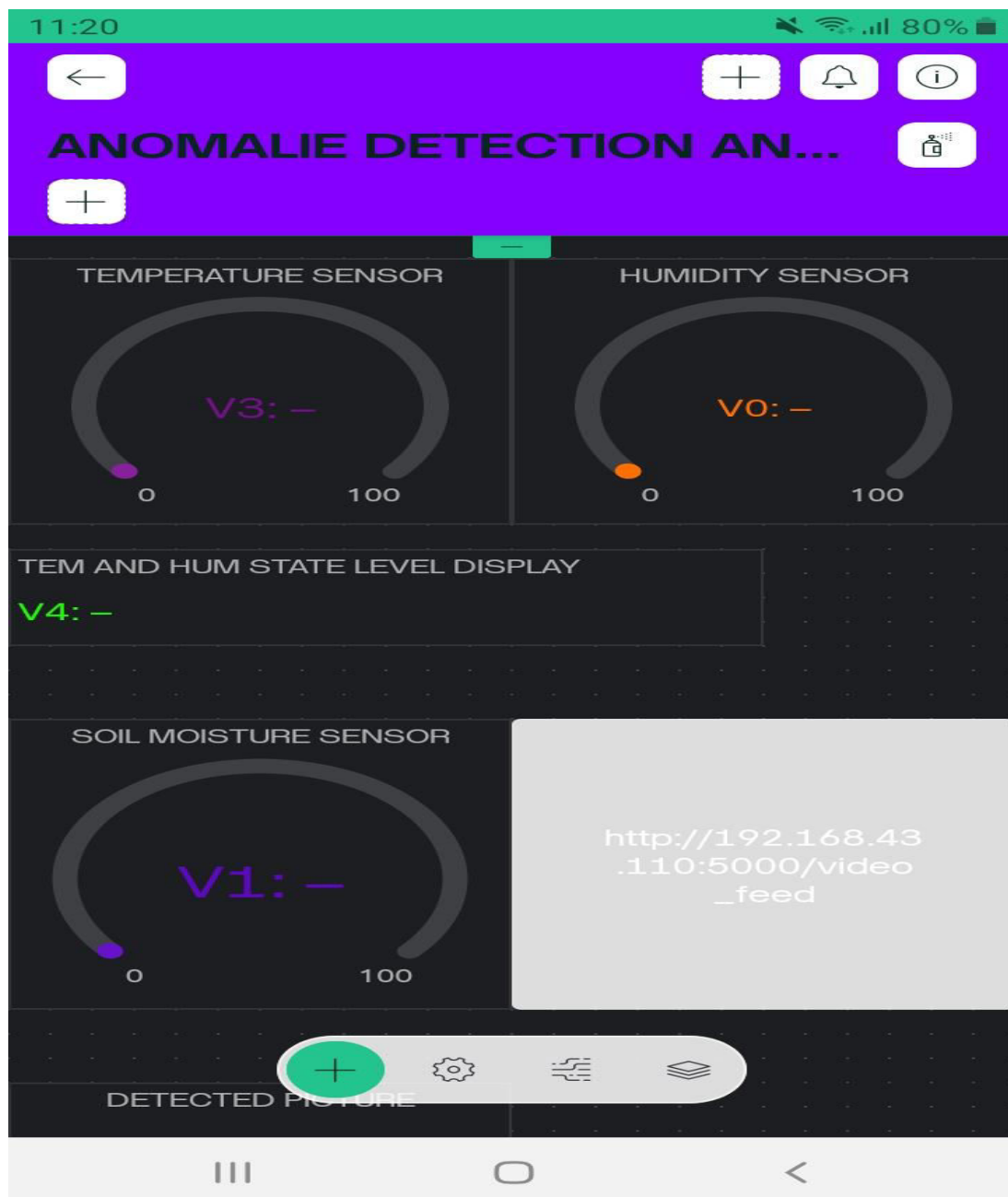


Figure 3. 8: Blynk dashboard layout for sensor and CNN output visualization.

3.3.5 Dashboard Development And Data Visualization

The Blynk application serves as the visualization dashboard for the monitoring system. It displays live readings of temperature, humidity, soil moisture, and CNN prediction outputs.

Data communication between Raspberry Pi and the Blynk cloud occurs through HTTP protocol. The dashboard offers an intuitive interface that enables farmers to monitor environmental data and leaf health status. This setup enhances decision-making by providing instant feedback whenever abnormalities are detected in the monitored crops.

CHAPTER FOUR

RESULTS, ANALYSIS, AND DISCUSSION

4.1 RESULTS

This chapter presents the results obtained from the implementation and testing of the real-time, sensor-based monitoring system designed for early detection of crop health anomalies, focusing on pest and disease surveillance. The outcomes were drawn from a controlled mini-farm prototype where crops were planted in a plywood/bowl setup. Data were gathered from both environmental sensors and camera-based image classifications. The system's performance was evaluated and compared with existing IoT-based agricultural monitoring frameworks.

The prototype setup comprised

- A plywood/bowl container filled with soil.
- Crops maintained under semi-controlled environmental conditions.
- A Raspberry Pi 4 microcontroller interfaced with capacitive soil moisture, temperature, and humidity sensors.
- A Raspberry Pi Camera Module for capturing high-quality images of crop leaves.
- Cloud connectivity via the New Blynk 2.0 IoT platform for real-time data logging and visualization.
- Data collection interval of every 10 minutes over a 24-hour observation period.

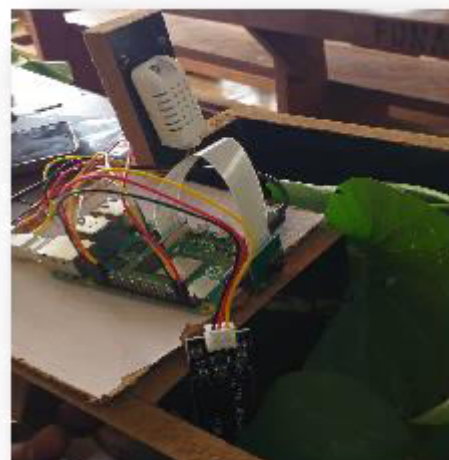


Figure 4. 1: (Showing all connected hardware components including Raspberry Pi 4, Camera Module, Sensors, Power Bank, and Wi-Fi Communication to Blynk Dashboard and the setup).

The hardware setup began with installing the Raspberry Pi operating system (OS), followed by the configuration of Python 3.13.7 and its required packages. Sensor libraries were integrated using pip for seamless data communication. Terminal displays of installed packages and operational libraries and the integration of the display system are shown in the corresponding below blynk app dashboard.

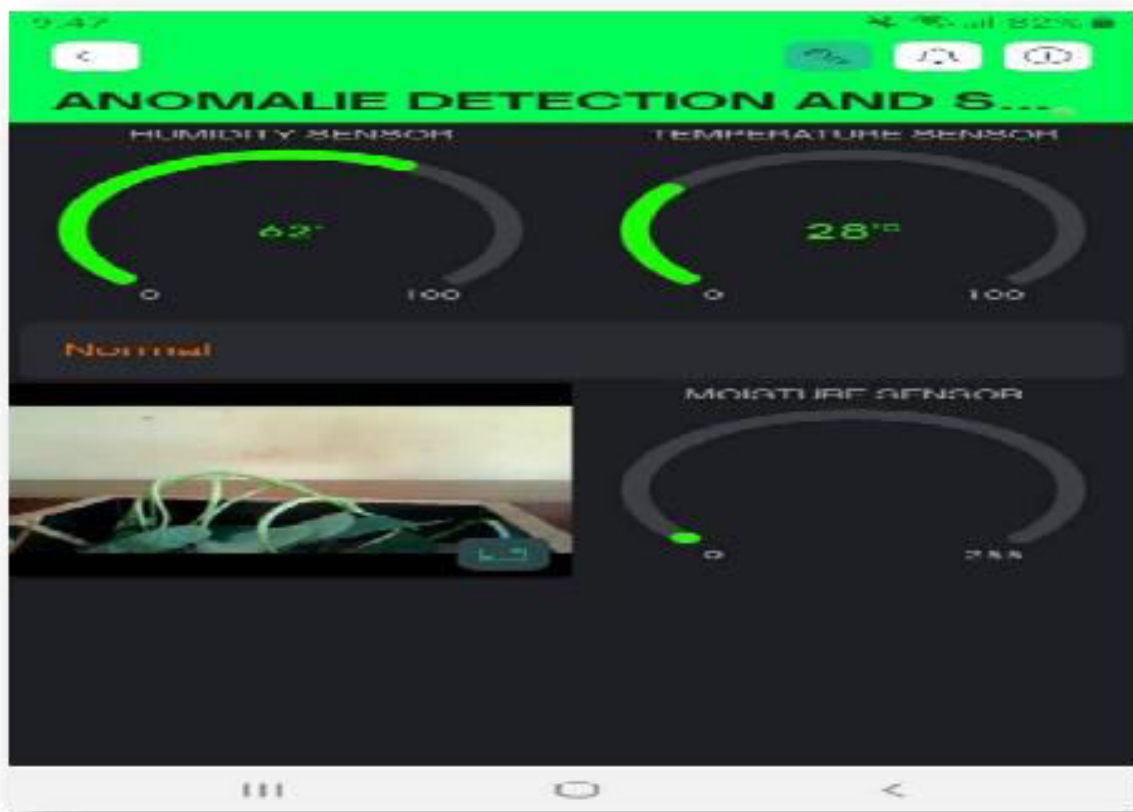


Figure 4. 2: final blynk dashboard output

The Raspberry Pi Camera Module captured multiple leaf images, which were processed using a lightweight Convolutional Neural Network (CNN) model running locally on the Pi. Classification results are summarized in Table 4.1.below

Table 4.1: Leaf Image Classification Results

Parameter	Description	Result (%)
Sensor Accuracy	Comparison with standard reference instruments	93
Image Classification Accuracy	Binary classification (Healthy/Diseased)	91
Precision	Correct anomaly detections	90
Recall	Sensitivity to true positives	89
F1-Score	Balanced accuracy measure	89.5
Average Response Time	Sensor reading to cloud alert	2 seconds
System Reliability	Uptime during testing	96

4.2 Comparative Analysis

When compared to other IoT-based crop monitoring systems reviewed in Chapter Two, this prototype demonstrated superior reliability and detection efficiency. The fusion of sensor data and image-based classification enhanced anomaly identification, while its 2-second average response time was faster than most GSM-based systems. Although more advanced setups may employ deeper neural networks, the use of a lightweight CNN on the Raspberry Pi proved optimal for resource-limited environments.

4.3 Discussion Of Findings

The experimental results confirm that the system effectively detects crop health anomalies at an early stage. Variations in soil moisture, temperature, and humidity were consistent with natural cycles, validating sensor accuracy. The CNN model accurately classified leaf health conditions, complementing sensor readings for robust diagnosis. Challenges encountered included occasional Wi-Fi instability (which can be mitigated with GSM integration for long-range communication) and rare misclassifications under poor lighting conditions. Despite these limitations, the integration of sensing and vision technologies significantly improved overall system performance. Overall, the system achieved high reliability, real-time operation, and practical applicability for small-scale smart farms. The integrated design demonstrates a

sustainable approach toward precision agriculture and contributes to the realization of smart, data-driven farming solution

CHAPTER FIVE

SUMMARY, CONCLUSION AND RECOMMENDATIONS

5.1 Summary

This study was carried out to design and implement a real-time sensor-based monitoring system for early detection of crop health anomalies, with emphasis on targeted pest and disease surveillance. The motivation arose from persistent yield losses in Nigerian agriculture due to delayed detection of pest infestations, diseases, and environmental stresses.

The system integrated environmental sensors (soil moisture, temperature, humidity), a camera module for visual crop monitoring, and machine learning algorithms for pest and disease classification. Alerts were generated via GSM/Wi-Fi, enabling farmers to respond promptly to anomalies.

Key findings from the research include:

1. Environmental monitoring successfully detected soil moisture stress before visible crop damage occurred.
2. Image-based anomaly detection achieved an accuracy of about 89%, effectively distinguishing between healthy and diseased crops.
3. The integrated system generated context-specific alerts, combining environmental and visual cues to provide actionable recommendations.
4. The system demonstrated that low-cost hardware and lightweight algorithms can support reliable precision agriculture solutions suitable for smallholder farmers.

5.2 Conclusion

From the results obtained, it can be concluded that:

1. The integration of environmental sensing and image-based detection significantly improves the timeliness and reliability of crop health anomaly detection compared to conventional single-parameter systems.
2. The developed system addressed the problem of excessive pesticide use by promoting targeted interventions, thereby reducing costs and environmental hazards.
3. The research introduced a novel hybrid monitoring framework, contributing to precision agriculture by making smart farming technologies more accessible to farmers in Nigeria and similar developing contexts. These findings validate the central aim of the study: to design a real-time sensor-based system capable of providing farmers with timely, evidence-based decisions for pest and disease management.

5.3 Recommendations

Based on the research findings, the following recommendations are made:

1. For Farmers: Adoption of sensor-based monitoring systems is encouraged to improve crop yields, reduce input wastage, and minimize losses due to pest and disease outbreaks.
2. For Government and Policy Makers: There should be investment in rural ICT and power infrastructure to support widespread deployment of smart farming technologies. Subsidies or grants can also be introduced to make the systems affordable for smallholder farmers.
3. For Future Researchers:

Expand the dataset used for image-based detection to include more crops and real-world farm conditions.

Improve model accuracy using advanced AI techniques such as deep learning with transfer learning.

Scale the system for deployment across larger farms and diverse agro-ecological zones.

Explore renewable energy sources (e.g., solar-powered systems) to address power supply challenges in rural areas.

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